Form B

Core Focus on Proportions & Probability

Block 4 Review ~ Probability and Random Sampling

ANSWER KEY

Name _____ Period _____ Date _____

Part I – Selected Response

1. Marcus planted 40 sunflower seeds and 28 of them sprouted. What is the experimental probability that the next sunflower seed he plants will sprout?

A. $\frac{3}{10}$ B. $\frac{7}{10}$ C. $\frac{2}{3}$ D. $\frac{4}{5}$

B ($\frac{28}{40} = \frac{7}{10}$)

2. A spinner has 3 yellow sections, 5 purple sections and 4 orange sections, all the same size. It is spun 36 times. For 2a – 2c, circle YES or NO to show whether you would expect the spinner to land on the given color the number of times stated.

2a. 12 times on yellow **YES NO**

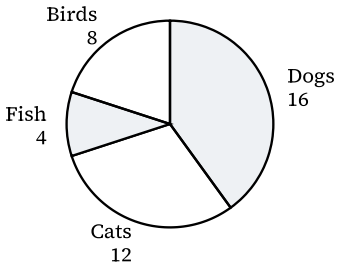
2b. 15 times on purple **YES NO**

2c. 12 times on orange **YES NO**

2a NO ($\frac{3}{12} \cdot 36 = 9$) · 2b YES ($\frac{5}{12} \cdot 36 = 15$) · 2c YES ($\frac{4}{12} \cdot 36 = 12$)

3. Nadia asked 40 students to name their favorite pet. The results are shown in the pie chart. If Nadia asked 300 students, how many would you predict would choose birds?

Favorite Pet



A. 8 students

B. 40 students

C. 60 students

D. 2,400 students

C ($\frac{8}{40} = \frac{1}{5}$; $\frac{1}{5} \cdot 300 = 60$)

4. A family has three children. Owen made a list of every possible order of boys and girls (for example, BGG). How many outcomes on his list have exactly two girls?

A. 2

B. 3

C. 4

D. 8

B (BGG, GBG, GGB out of 8 outcomes)

5. You flip a coin and then roll a number cube. Which of the theoretical probabilities are accurate? Circle all that apply.

A. P(heads then 4) = $\frac{1}{12}$

B. P(tails then even number) = $\frac{1}{4}$

C. P(heads then 7) = $\frac{1}{12}$

D. P(tails then number less than 3) = $\frac{1}{3}$

E. P(tails then number greater than 4) = $\frac{1}{6}$

A, B, E. (A: ½ · ⅙ = $\frac{1}{12}$; B: ½ · ½ = ¼; C is 0, a cube has no 7; D: ½ · $\frac{2}{6} = \frac{1}{6}$, not ⅓; E: ½ · $\frac{2}{6} = \frac{1}{6}$)

6. Eight students are running for class president and vice president. In how many different ways can these two offices be filled?

A. 15 B. 16 C. 56 D. 64

C (8 choices for president · 7 left for vice president = 56)

7. A gym locker has a four-digit code. Each digit in the code must be a 1, 2 or 3. What is the probability the code is 1111?

A. $\frac{1}{81}$

B. $\frac{1}{64}$

C. $\frac{1}{12}$

D. $\frac{1}{3}$

A (3 · 3 · 3 · 3 = 81 codes, only one is 1111)

8. The cafeteria manager wants to know which new lunch item the students at Ridge Middle School would like most. Which sample will give the best chance of an accurate prediction?

A survey of 60 students _____.

A. in the cooking club.

B. chosen by drawing 60 names from a list of all students at the school.

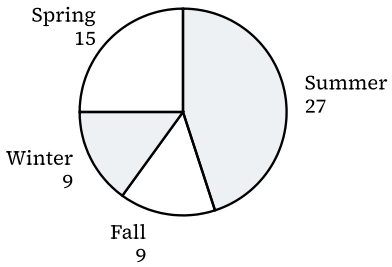
C. who are first to drop a form in the suggestion box.

D. in the eighth grade.

B (a random sample of the whole school; the others leave out groups or only include volunteers)

9. Sixty students were surveyed about their favorite season. The data is below. Which inference is the weakest?

Favorite Season



A. Summer is the most popular season.

B. Fall and winter are equally popular.

C. More students prefer spring than winter.

D. More than half of the students prefer summer.

D (summer is 27 of 60 = 45%, less than half)

10. The five-number summary and mean for two data sets are shown below. Each data set has 11 values. Which data set has the smaller mean absolute deviation?

Set P: 30 ~ 31 ~ 33 ~ 34 ~ 36
mean = 33

Set Q: 10 ~ 20 ~ 32 ~ 45 ~ 58
mean = 33

A. Set P

B. Set Q

C. Set P and Set Q have equal mean absolute deviations

D. Cannot be determined

A (Set P's values all lie within 3 of the mean, so they are much less spread out)

11. The points Ava and Ben scored in seven basketball games are shown below. Which statement best compares their median scores?

Ava: 12, 9, 15, 10, 14, 9, 11
Ben: 8, 13, 16, 10, 13, 12, 7

A. Ava's median is 1 point greater than Ben's.

B. Ben's median is 1 point greater than Ava's.

C. Their medians differ by 2 points.

D. Their medians are equal.

B (in order, Ava: 9, 9, 10, 11, 12, 14, 15; Ben: 7, 8, 10, 12, 13, 13, 16)

Form B

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Part II – Constructed Response

Theo has 30 tiles numbered 1 to 30 in a bag. He randomly draws one tile. Find each probability for the tile he draws.

1. P(12)

$\frac{1}{30}$

2. P(multiple of 5)

$\frac{6}{30} = \frac{1}{5}$ (5, 10, 15, 20, 25, 30)

3. P(35)

0 (impossible, there is no tile 35)

4. P(not a multiple of 3)

$\frac{20}{30} = \frac{2}{3}$ (10 tiles are multiples of 3)

5. A box holds 8 red, 4 yellow and 12 green markers. Ella reaches in and takes one marker without looking. Which color has a $\frac{1}{3}$ chance of being chosen? Explain how you know your answer is correct.

Red. There are 8 + 4 + 12 = 24 markers and P(red) = $\frac{8}{24} = \frac{1}{3}$. (Yellow is $\frac{4}{24} = \frac{1}{6}$ and green is $\frac{12}{24} = \frac{1}{2}$.)

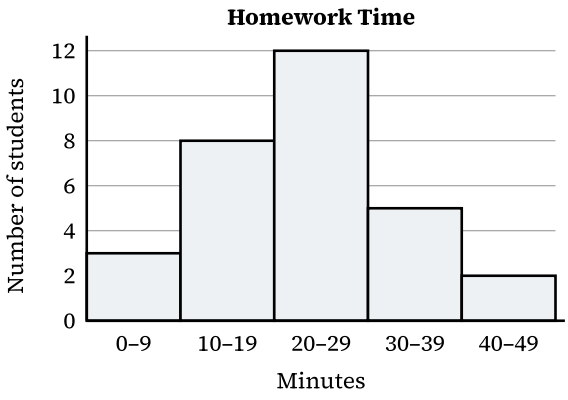
6. A biologist wanted to estimate the number of turtles in a pond. She caught 12 turtles, tagged them and let them go. A week later she caught 30 turtles, and 4 of them had tags. Estimate the number of turtles in the pond. Show all work necessary to justify your answer.

90 turtles. $\frac{\text{4 tagged}}{\text{30 caught}} = \frac{\text{12 tagged}}{n \text{ total}}$; $4n = 360$, $n = 90$. (Check: 4 of 30 is $\frac{2}{15}$, and $\frac{12}{90} = \frac{2}{15}$.)

7. Rosa rolled a regular number cube 54 times. How many times would you predict she rolled a number less than 3? Explain your reasoning.

18 times. Two of the six numbers (1 and 2) are less than 3, so P = $\frac{2}{6} = \frac{1}{3}$, and $\frac{1}{3} \cdot 54 = 18$.

8. The histogram shows the number of minutes students in a class spent on homework last night.



a. How many students are in the class?

3 + 8 + 12 + 5 + 2 = 30 students

b. Find the probability that a randomly chosen student spent 20 to 29 minutes on homework.

$\frac{12}{30} = \frac{2}{5}$

9. A sandwich shop offers three breads (white, wheat and rye) and four fillings (turkey, ham, tuna and cheese). Each sandwich has one bread and one filling. How many different sandwiches are possible? Show a tree diagram, list or table to support your answer.

12 sandwiches (3 · 4). List: white–turkey, white–ham, white–tuna, white–cheese, wheat–turkey, wheat–ham, wheat–tuna, wheat–cheese, rye–turkey, rye–ham, rye–tuna, rye–cheese.

10. Jamal takes a quiz with four questions. Each question has four possible answers. If Jamal guesses on every question, what is the probability he answers them all correctly?

$\frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{256}$

11. Devon spins a spinner with five equal sections numbered 1 – 5. He then draws a card from ten cards numbered 1 – 10. What is the probability Devon spins a 4 and draws an even number?

$\frac{1}{5} \cdot \frac{5}{10} = \frac{5}{50} = \frac{1}{10}$

12. The school library wants to know what type of book students at the school like best. Mr. Ortiz surveys the 25 students at a meeting of the graphic novel club. Most of them say graphic novels, so he decides graphic novels are the favorite of most students at the school. Is his prediction accurate based on the sample surveyed? Explain your reasoning.

No. The sample is biased: members of a graphic novel club are much more likely to choose graphic novels than other students. He should survey a random sample of students from the whole school.

13. Two different random samples were used to find students' favorite school lunch. The results are below. Make at least two inferences.

	Pizza	**Tacos**	**Salad**	**Pasta**	**TOTAL**
Sample 1	24	18	6	12	60
Sample 2	35	15	10	10	70

Sample answers: Pizza is the most popular lunch (24 of 60 = 40% and 35 of 70 = 50%). Salad is one of the least popular (10% and about 14%). Tacos are the second favorite in both samples (30% and about 21%).

Use the data below to answer numbers 14 – 20. The data shows the number of books 10 students in each class read over the summer.

Class A: 8, 2, 12, 5, 9, 4, 12, 6, 5, 7
Class B: 9, 6, 10, 8, 7, 9, 11, 8, 12, 10

14. Find the mean of each data set.

Class A: $\frac{70}{10}$ = 7 books. Class B: $\frac{90}{10}$ = 9 books.

15. Find the median of each data set.

Class A: 2, 4, 5, 5, 6, 7, 8, 9, 12, 12 → 6.5. Class B: 6, 7, 8, 8, 9, 9, 10, 10, 11, 12 → 9.

16. Jada said that, on average (mean), Class B read 2 more books than Class A. Is her comparison accurate? Explain your reasoning.

Yes. The means are 9 and 7, and 9 − 7 = 2 books.

17. Find the five-number summary of each data set.

Class A: 2 ~ 5 ~ 6.5 ~ 9 ~ 12. Class B: 6 ~ 8 ~ 9 ~ 10 ~ 12. (Q1 and Q3 are the medians of the lower 5 and upper 5 values.)

18. Find the interquartile range of each data set.

Class A: 9 − 5 = 4. Class B: 10 − 8 = 2.

19. Find the mean absolute deviation of each data set.

Class A (mean 7): distances 5, 3, 2, 2, 1, 0, 1, 2, 5, 5; sum 26; MAD = $\frac{26}{10}$ = 2.6 books. Class B (mean 9): distances 3, 2, 1, 1, 0, 0, 1, 1, 2, 3; sum 14; MAD = $\frac{14}{10}$ = 1.4 books.

20. Write one sentence comparing a measure of variability between the two data sets.

Sample: Class A has a larger MAD (2.6 vs. 1.4), so the number of books read in Class A is more spread out. (Or: Class A's IQR of 4 is twice Class B's IQR of 2.)

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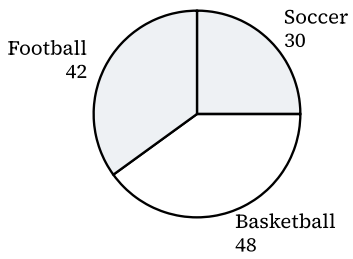
ANSWER KEY

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Part III – Problem Solving

Maya asked a random sample of 120 students at her school which sport they most like to watch and which drink they most like to have while watching. The results are below.

Favorite Sport to Watch



Favorite Drink	Number of Students
Water	60
Juice	36
Milk	24

There are 500 students at Maya's school. Based on Maya's data, what is the probability that the next student she asks will prefer watching basketball and drinking juice? How many students in the whole school would you predict prefer both basketball and juice?

P(basketball) = $\frac{48}{120} = \frac{2}{5}$. P(juice) = $\frac{36}{120} = \frac{3}{10}$. P(basketball and juice) = $\frac{2}{5} \cdot \frac{3}{10} = \frac{6}{50} = \frac{3}{25}$. Prediction: $\frac{3}{25} \cdot 500 = 60$ students.